STAT 443 Project

```r
library(forecast)
```

```
Warning: package 'forecast' was built under R version 4.4.3
```

```r
# Function for variogram as defined in
# Bisgaard and Kulahci (2011). Time Series Analysis and Forecasting by Example, Wiley
variogram = function(y, lagmax=10, iprint=F)
{ G = rep(1,lagmax)
n = length(y)
if(lagmax>n) { lagmax = n-2 }
y1 = y[-1]; y2 = y[-n]
d1 = y1-y2; denom = var(d1)
for(k in 2:lagmax)
{ y1 = y[(k+1):n]; y2 = y[1:(n-k)]
dk = y1-y2
numer = var(dk)
G[k] = numer/denom
}
# H is the variogram assuming a stationary time series
H = rep(1,lagmax)
ac = c(acf(y,plot=F,lag.max=lagmax)$acf)
H = (1-ac[-1])/(1-ac[2])
if(iprint)
{ print(cbind(G,H))
mx = max(cbind(G,H))
matplot(1:lagmax,cbind(G,H),ylim=c(0,mx),ylab="variogram", xlab="lag")
abline(h=1)
}
list(G=G, H=H)
}
```

```r
persist_fc <- function(train, holdout, iprint = FALSE) {
  ntrain <- length(train)
  nh <- length(holdout)
  sse <- 0
  fc_vec <- numeric(nh)
  fc <- train[ntrain]
  yt <- holdout[1]
  fc_vec[1] <- fc
  fcerror <- yt - fc
  sse <- sse + (fcerror^2)
  for (i in 2:nh) {
    yt <- holdout[i]
    fc <- holdout[i-1]
    fc_vec[i] <- fc
    fcerror <- yt - fc
    sse <- sse + (fcerror^2)
  }
  return(list(rmse = sqrt(sse/nh), fc = fc_vec))
}

# Linear in the previous observation
# b0,b1 from least squares or another estimation method
linreg_fc = function(train_last, holdout, b0,b1)
{ nh = length(holdout)
  previous = c(train_last, holdout[-nh])
  fc_vec = b0+b1*previous
  rmse = sqrt(mean((holdout-fc_vec)^2))
  return(list(rmse=rmse, fc=fc_vec))
}

# Simple Exponential Smoothing
esm_fc <- function(train, holdout, alpha, level, iprint = FALSE) {
  nh <- length(holdout)
  sse <- 0
  fc_vec <- numeric(nh)
  L <- level
  for (i in 1:nh) {
    fc <- L
    fc_vec[i] <- fc
    yt <- holdout[i]
    fcerror <- yt - fc
    sse <- sse + (fcerror^2)
```

```r
    L <- alpha * yt + (1 - alpha) * L
  }
  return(list(rmse = sqrt(sse / nh), fc = fc_vec))
}
```

```r
plot_forecast <- function(holdout, fc, model_name = "Model") {

  x <- seq_along(holdout)

  plot(x, holdout, type = "l", col = "black", lwd = 2,
       main = paste("Forecast vs Actual -", model_name),
       ylab = "Value", xlab = "Time")

  lines(x, fc, col = "red", lwd = 2)

  legend("topleft",
         legend = c("Actual", "Forecast"),
         col = c("black", "red"),
         lty = 1, lwd = 2)
}
```

```r
arima_rolling_fc <- function(fitted_model, holdout, iprint = FALSE) {
  nh <- length(holdout)
  sse <- 0
  fc_vec <- numeric(nh)

  # Extract the training data from the fitted model
  train_data <- fitted_model$x
  current_data <- train_data

  for (i in 1:nh) {
    # Forecast one step ahead using current model
    fc <- forecast(Arima(current_data, model = fitted_model), h = 1)$mean[1]
    fc_vec[i] <- fc

    yt <- holdout[i]
    fcerror <- yt - fc
    sse <- sse + (fcerror^2)

    # Expand training data with actual observed value
    current_data <- c(current_data, yt)
  }
```

```r
  return(list(rmse = sqrt(sse / nh), fc = fc_vec))
}

rolling_armax_fixed <- function(y_train, x_train, y_holdout, x_holdout,
                                order, include_mean = TRUE) {
  nh <- length(y_holdout)
  fc_vec <- numeric(nh)

  y_hist <- y_train
  x_hist <- as.matrix(x_train)
  x_holdout <- as.matrix(x_holdout)

  for (i in 1:nh) {
    fit <- forecast::Arima(
      y_hist,
      order = order,
      xreg = x_hist,
      include.mean = include_mean,
      method = "ML"
    )

    newx <- matrix(x_holdout[i, ], nrow = 1)

    fc_vec[i] <- forecast::forecast(
      fit,
      xreg = newx,
      h = 1
    )$mean[1]

    y_hist <- c(y_hist, y_holdout[i])
    x_hist <- rbind(x_hist, newx)
  }

  rmse <- sqrt(mean((y_holdout - fc_vec)^2))
  list(rmse = rmse, fc = fc_vec)
}
```

```r
rolling_armax <- function(y_train, x_train, y_holdout, x_holdout,
                          order = c(0,0,1), include_mean = TRUE) {
  nh <- length(y_holdout)
  fc_vec <- numeric(nh)
```

```r
  y_hist <- as.numeric(y_train)
  x_hist <- as.matrix(x_train)
  x_holdout <- as.matrix(x_holdout)

  for (i in 1:nh) {
    fit <- forecast::Arima(
      y_hist,
      order = order,
      xreg = x_hist,
      include.mean = include_mean,
      method = "ML"
    )

    newx <- matrix(x_holdout[i, ], nrow = 1)

    fc_vec[i] <- as.numeric(
      forecast::forecast(fit, xreg = newx, h = 1)$mean
    )

    # update history with the ACTUAL next observation
    y_hist <- c(y_hist, y_holdout[i])
    x_hist <- rbind(x_hist, newx)
  }

  rmse <- sqrt(mean((y_holdout - fc_vec)^2))
  list(rmse = rmse, fc = fc_vec)
}

armax_fc = function(tsdata, ntrain, order, traincoef, xreg)
{
  coef_names <- names(traincoef)

  include_mean <- any(coef_names %in% c("intercept", "mean"))

  obj = arima(tsdata,
              order = order,
              init = traincoef,
              fixed = traincoef,
              xreg = xreg,
              include.mean = include_mean,
              method = "ML",
              optim.control = list(maxit = 0))
```

```r
  fc = tsdata - obj$residuals

  holdout_fc = fc[(ntrain+1):length(tsdata)]
  holdout = tsdata[(ntrain+1):length(tsdata)]

  rmse = sqrt(mean((holdout - holdout_fc)^2))

  return(list(rmse = rmse, fc = holdout_fc))
}
```

Load Data

```r
# Load Spot Crude Oil Price
#crude_oil <- read.csv("data/WTISPLC.csv")
#crude_oil_ts <- ts(crude_oil$WTISPLC,start=c(1975,1,1),freq=12)

#head(crude_oil_ts)

# Load Spot Crude Oil Price
crude_oil <- read.csv("data/bloomberg_oil_spot_monthly.tsv", header = TRUE, sep = "\t")
crude_oil$Date <- as.Date(crude_oil$Date)
crude_oil <- crude_oil[order(crude_oil$Date), ]

crude_oil_ts <- ts(
  crude_oil$Last.Price,
  start = c(as.numeric(format(min(crude_oil$Date), "%Y")),
            as.numeric(format(min(crude_oil$Date), "%m"))),
  frequency = 12
)

head(crude_oil_ts)
```

```
     Jan Feb Mar Apr May Jun
1970 1.8 1.8 1.8 1.8 1.8 1.8
```

```r
#write.table(crude_oil_ts, "savedData/crude_oil_ts_new.csv", sep = ",", row.names = FALSE)
```

Split Training - Holdout

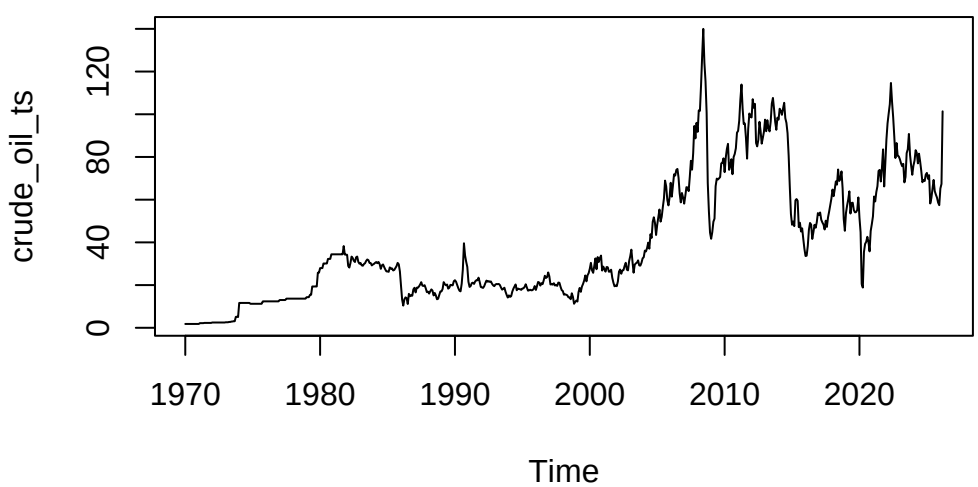
```
crude_oil_train_ts = window(crude_oil_ts, start = c(1975,1), end = c(2010,12))
crude_oil_holdout_ts = window(crude_oil_ts, start = c(2011,1), end = c(2026,1))

d_crude_oil_train_ts = diff(crude_oil_train_ts)
d_crude_oil_holdout_ts = diff(crude_oil_holdout_ts)

dl_crude_oil_ts <- diff(log(crude_oil_ts))
dl_crude_oil_train_ts = diff(log(crude_oil_train_ts))
dl_crude_oil_holdout_ts = diff(log(crude_oil_holdout_ts))
```

```
end(d_crude_oil_holdout_ts)
```

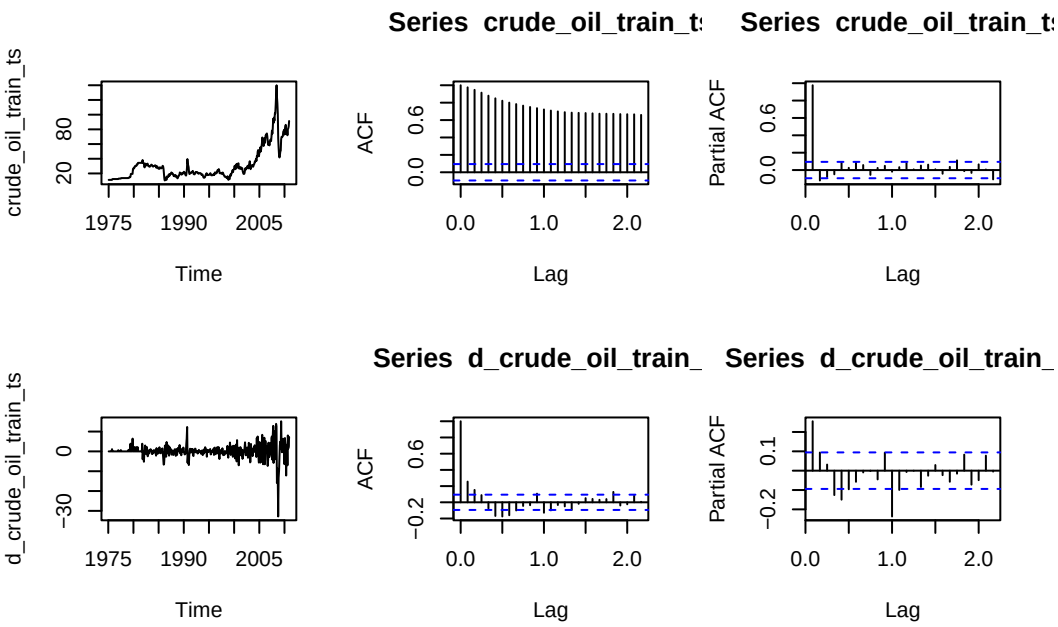
```
[1] 2026    1
```

##Diagnostic Plots

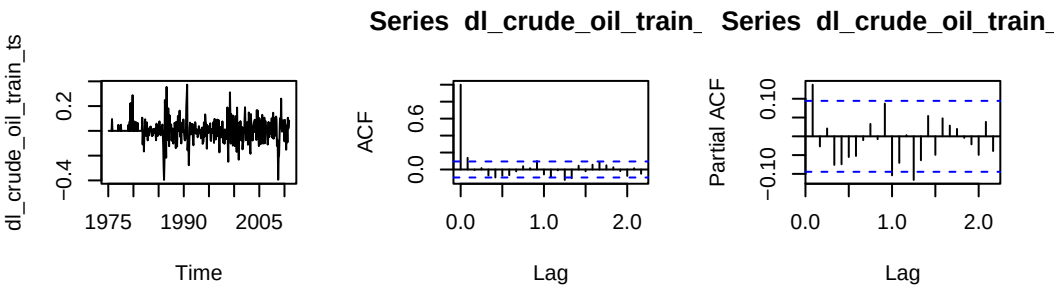
```
plot(crude_oil_ts)
```



```r
par(mfrow = c(2,3))

plot.ts(crude_oil_train_ts)
acf(crude_oil_train_ts)
pacf(crude_oil_train_ts)

plot.ts(d_crude_oil_train_ts)
acf(d_crude_oil_train_ts)
pacf(d_crude_oil_train_ts)
```



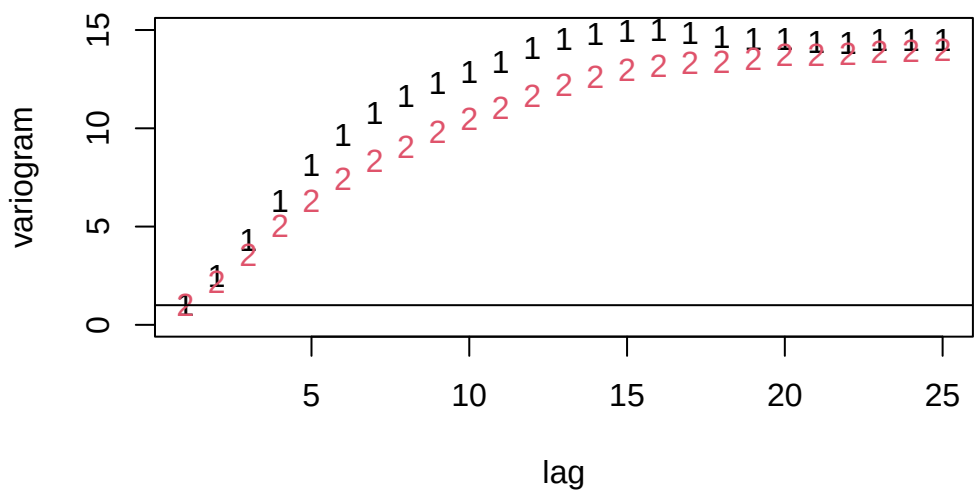

```r
plot.ts(dl_crude_oil_train_ts)
acf(dl_crude_oil_train_ts)
pacf(dl_crude_oil_train_ts)
```




```r
variogram(crude_oil_train_ts, lagmax=25, iprint=T)
```

```
              G         H
 [1,]  1.000000  1.000000
 [2,]  2.507923  2.213314
 [3,]  4.322784  3.575911
 [4,]  6.320108  5.027343
 [5,]  8.134092  6.293844
 [6,]  9.654462  7.451824
 [7,] 10.812447  8.365963
 [8,] 11.642819  9.074476
 [9,] 12.316291  9.817642
[10,] 12.897735 10.476988
[11,] 13.397260 11.046448
[12,] 14.086429 11.660209
[13,] 14.545171 12.204713
[14,] 14.793300 12.609311
[15,] 14.972980 12.973769
[16,] 15.014611 13.202572
[17,] 14.856925 13.314807
[18,] 14.657656 13.401482
[19,] 14.568813 13.556345
```

```
[20,] 14.541040 13.718163
[21,] 14.416240 13.731384
[22,] 14.355121 13.774623
[23,] 14.491999 13.909540
[24,] 14.497082 13.961572
[25,] 14.500442 14.019781
```

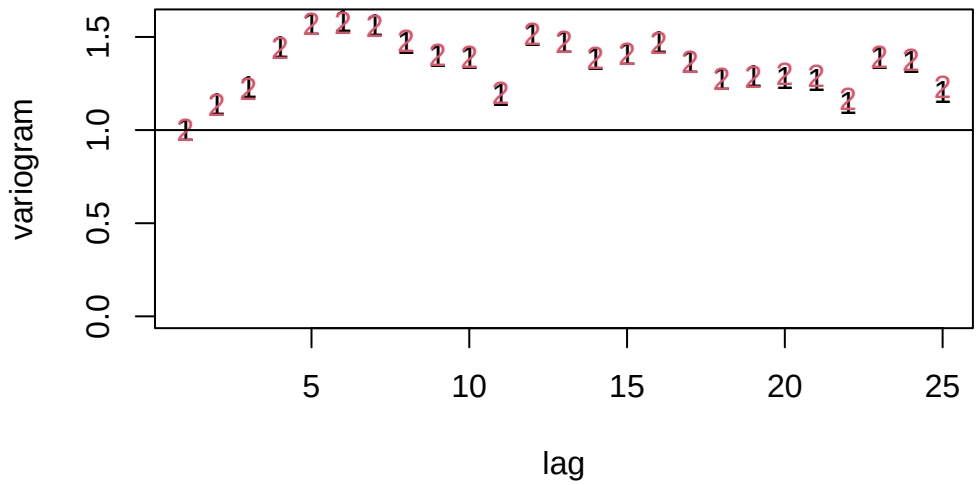



```
$G
 [1]  1.000000  2.507923  4.322784  6.320108  8.134092  9.654462 10.812447
 [8] 11.642819 12.316291 12.897735 13.397260 14.086429 14.545171 14.793300
[15] 14.972980 15.014611 14.856925 14.657656 14.568813 14.541040 14.416240
[22] 14.355121 14.491999 14.497082 14.500442

$H
 [1]  1.000000  2.213314  3.575911  5.027343  6.293844  7.451824  8.365963
 [8]  9.074476  9.817642 10.476988 11.046448 11.660209 12.204713 12.609311
[15] 12.973769 13.202572 13.314807 13.401482 13.556345 13.718163 13.731384
[22] 13.774623 13.909540 13.961572 14.019781
```

```r
variogram(d_crude_oil_train_ts, lagmax=25, iprint=T)
```

```
              G         H
```

```
 [1,] 1.000000 1.000000
 [2,] 1.140406 1.137681
 [3,] 1.230013 1.223906
 [4,] 1.448366 1.443887
 [5,] 1.572692 1.568983
 [6,] 1.584348 1.577965
 [7,] 1.567000 1.557561
 [8,] 1.471277 1.478062
 [9,] 1.397354 1.403207
[10,] 1.386399 1.391114
[11,] 1.187448 1.199834
[12,] 1.511651 1.516234
[13,] 1.473859 1.476647
[14,] 1.385325 1.387892
[15,] 1.407840 1.410880
[16,] 1.472604 1.469787
[17,] 1.369459 1.367682
[18,] 1.274489 1.273881
[19,] 1.290106 1.287193
[20,] 1.279969 1.301071
[21,] 1.271118 1.289944
[22,] 1.145389 1.170839
[23,] 1.385722 1.395920
[24,] 1.366577 1.375749
[25,] 1.207327 1.234587
```

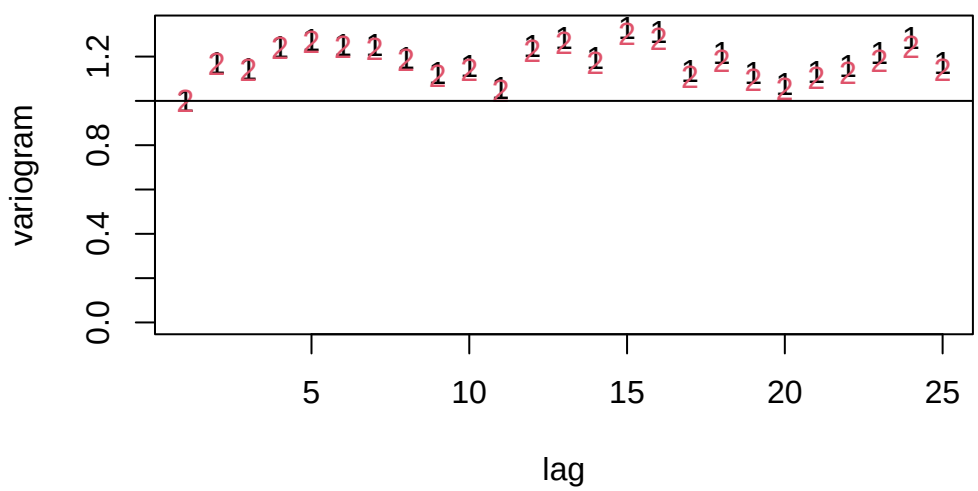



```
$G
 [1] 1.000000 1.140406 1.230013 1.448366 1.572692 1.584348 1.567000 1.471277
 [9] 1.397354 1.386399 1.187448 1.511651 1.473859 1.385325 1.407840 1.472604
[17] 1.369459 1.274489 1.290106 1.279969 1.271118 1.145389 1.385722 1.366577
[25] 1.207327

$H
 [1] 1.000000 1.137681 1.223906 1.443887 1.568983 1.577965 1.557561 1.478062
 [9] 1.403207 1.391114 1.199834 1.516234 1.476647 1.387892 1.410880 1.469787
[17] 1.367682 1.273881 1.287193 1.301071 1.289944 1.170839 1.395920 1.375749
[25] 1.234587
```

```r
variogram(dl_crude_oil_train_ts, lagmax=25, iprint=T)
```

```
             G        H
 [1,] 1.000000 1.000000
 [2,] 1.170684 1.167915
 [3,] 1.146258 1.140938
 [4,] 1.246487 1.239469
 [5,] 1.276368 1.267777
 [6,] 1.255360 1.244322
 [7,] 1.251255 1.237415
```

```
 [8,] 1.196123 1.184554
 [9,] 1.126131 1.114659
[10,] 1.156505 1.142099
[11,] 1.057318 1.043851
[12,] 1.246984 1.227380
[13,] 1.285195 1.261666
[14,] 1.194501 1.170825
[15,] 1.331962 1.302037
[16,] 1.310576 1.278328
[17,] 1.136224 1.107609
[18,] 1.217635 1.183091
[19,] 1.127386 1.094287
[20,] 1.078727 1.056457
[21,] 1.129810 1.102712
[22,] 1.158570 1.129139
[23,] 1.215641 1.181328
[24,] 1.286229 1.246404
[25,] 1.171995 1.142738
```




```
$G
 [1] 1.000000 1.170684 1.146258 1.246487 1.276368 1.255360 1.251255 1.196123
 [9] 1.126131 1.156505 1.057318 1.246984 1.285195 1.194501 1.331962 1.310576
```

```
[17] 1.136224 1.217635 1.127386 1.078727 1.129810 1.158570 1.215641 1.286229
[25] 1.171995

$H
 [1] 1.000000 1.167915 1.140938 1.239469 1.267777 1.244322 1.237415 1.184554
 [9] 1.114659 1.142099 1.043851 1.227380 1.261666 1.170825 1.302037 1.278328
[17] 1.107609 1.183091 1.094287 1.056457 1.102712 1.129139 1.181328 1.246404
[25] 1.142738
```

We first try ARIMA

ARMA on difference

```r
dwti_arma = auto.arima(d_crude_oil_train_ts, seasonal=F, stationary=T)
print(dwti_arma)
```

```
Series: d_crude_oil_train_ts
ARIMA(2,0,2) with zero mean

Coefficients:
         ar1      ar2      ma1     ma2
      1.4117  -0.7988  -1.2275  0.7268
s.e.  0.0819   0.0655   0.0823  0.0733

sigma^2 = 11.95:  log likelihood = -1144.38
AIC=2298.75   AICc=2298.9   BIC=2319.09
```

Rolling 1-step forward

```r
dwti_arma_rolling = arima_rolling_fc(dwti_arma, d_crude_oil_holdout_ts)
dwti_arma_rolling$rmse
```

```
[1] 6.550004
```

ARMA on log return

```r
dlwti_arma = auto.arima(dl_crude_oil_train_ts, seasonal=F, stationary=T)
print(dlwti_arma)
```

```
Series: dl_crude_oil_train_ts
ARIMA(0,0,1) with zero mean

Coefficients:
         ma1
      0.1459
s.e.  0.0486

sigma^2 = 0.007729:  log likelihood = 436.85
AIC=-869.7   AICc=-869.68   BIC=-861.57
```

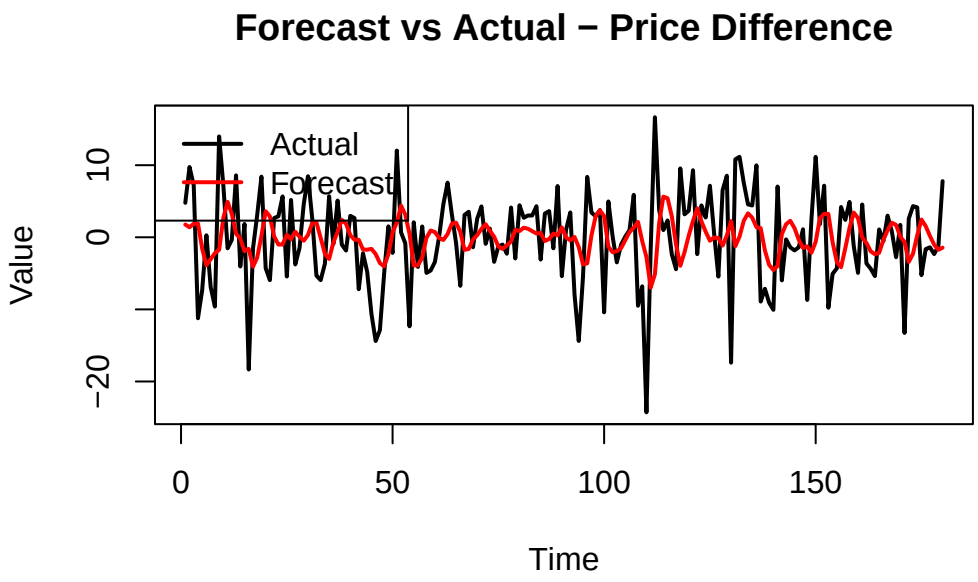
Rolling 1-step forward

```r
dlwti_arma_rolling = arima_rolling_fc(dlwti_arma, dl_crude_oil_holdout_ts)
dlwti_arma_rolling$rmse
```

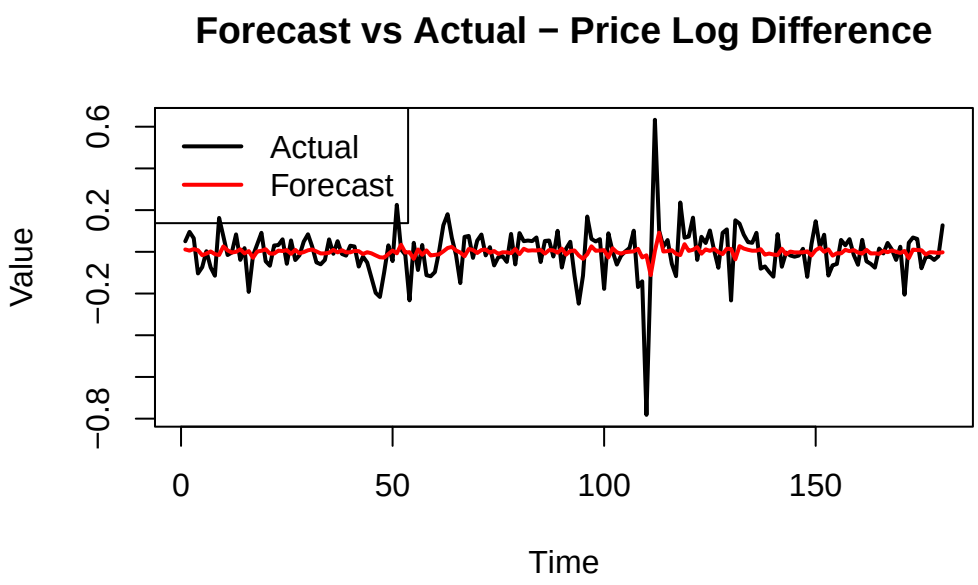
```
[1] 0.1134567
```

Plots

```r
plot_forecast(d_crude_oil_holdout_ts, dwti_arma_rolling$fc, "Price Difference")
```




```
plot_forecast(dl_crude_oil_holdout_ts, dlwti_arma_rolling$fc, "Price Log Difference")
```



We then introduce extrogenous variables and split Train-Holdout

```
#usd_idx = read.csv("data/TWEXBGSMTH.csv")
#usd_idx = read.csv("data/dxy_monthly.csv")
#ind_prod = read.csv("data/INDPRO.csv")
#opec_prod = read.csv("data/opcrtotl_monthly.tsv")

#usd index
usd_idx <- read.csv("data/dxy_monthly.tsv", header = TRUE, sep = "\t")
usd_idx$Date <- as.Date(usd_idx$Date)
usd_idx <- usd_idx[order(usd_idx$Date), ]

usd_idx_ts <- ts(
  usd_idx$Last.Price,
  start = c(1970,04),
  end = c(2026,04),
  frequency = 12
)
usd_idx_train_ts = window(usd_idx_ts, start = c(1975,1), end = c(2010,12))
usd_idx_holdout_ts = window(usd_idx_ts, start = c(2011,1), end = c(2026,1))

#Industrial production
ind_prod = read.csv("data/INDPRO.csv")
ind_prod$observation_date <- as.Date(ind_prod$observation_date)
ind_prod_ts <- ts(ind_prod$INDPRO, start = c(1919,1), end = c(2026,2), frequency = 12)
ind_prod_train_ts = window(ind_prod_ts, start = c(1975,1), end = c(2010,12))
ind_prod_holdout_ts = window(ind_prod_ts, start = c(2011,1), end = c(2026,1))

# OPEC oil production
opec_prod <- read.csv("data/opcrtotl_monthly.tsv", header = TRUE, sep = "\t")
opec_prod$Date <- as.Date(opec_prod$Date)
opec_prod <- opec_prod[order(opec_prod$Date), ]

opec_prod_ts <- ts(
  opec_prod$Last.Price,
  start = c(1970,12),
  end = c(2026,03),
  frequency = 12
)
opec_prod_train_ts = window(opec_prod_ts, start = c(1975,1), end = c(2010,12))
opec_prod_holdout_ts = window(opec_prod_ts, start = c(2011,1), end = c(2026,1))
```

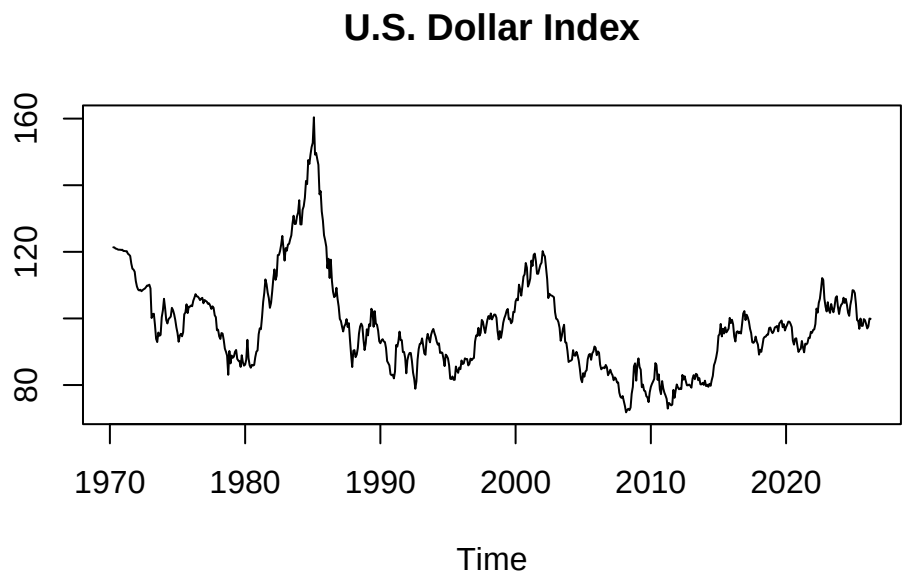
```r
# World MSCI Oil/Energy Company Index
#TO DO



head(usd_idx)
```

```
          Date Last.Price
673 1970-04-30    121.360
672 1970-05-29    121.322
671 1970-06-30    120.978
670 1970-07-31    120.904
669 1970-08-31    120.704
668 1970-09-30    120.620
```

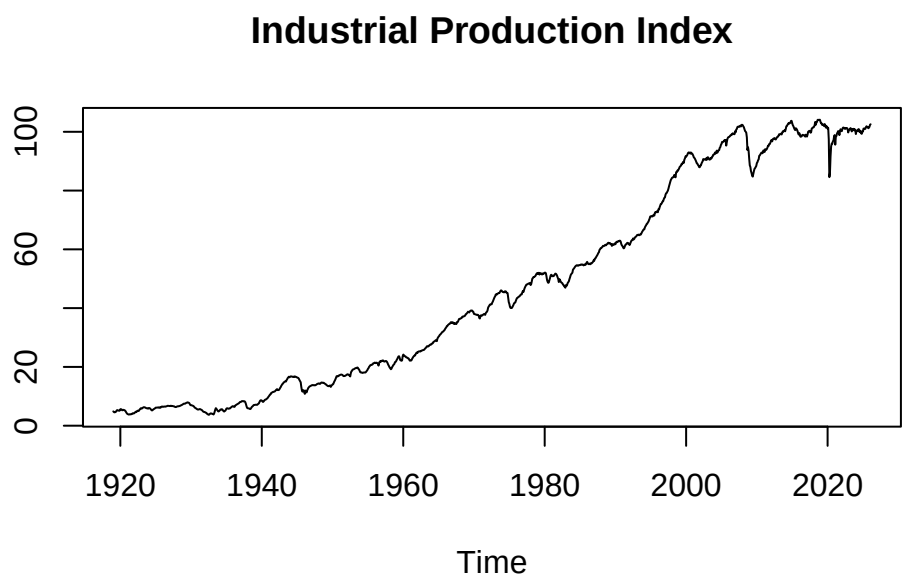
```r
#Saving data
#write.table(usd_idx_train_ts, "savedData/usd_idx_train_ts_new.csv", sep = ",", row.names = F
#write.table(usd_idx_holdout_ts, "savedData/usd_idx_holdout_ts_new.csv", sep = ",", row.names

#write.table(ind_prod_train_ts, "savedData/ind_prod_train_ts_new.csv", sep = ",", row.names =
#write.table(ind_prod_holdout_ts, "savedData/ind_prod_holdout_ts_new.csv", sep = ",", row.nam

#write.table(opec_prod_train_ts, "savedData/opec_prod_train_ts_new.csv", sep = ",", row.names
#write.table(opec_prod_holdout_ts, "savedData/opec_prod_holdout_ts_new.csv", sep = ",", row.r
```

```r
#usd_ts <- ts(usd_idx$Last.Price, start = c(1975,1), frequency = 12)
#ind_ts <- ts(ind_prod$INDPRO, start = c(1975,1), frequency = 12)
#opec_ts <- ts(opec_prod$Last.Price, start = c(1975,1), frequency = 12)
```

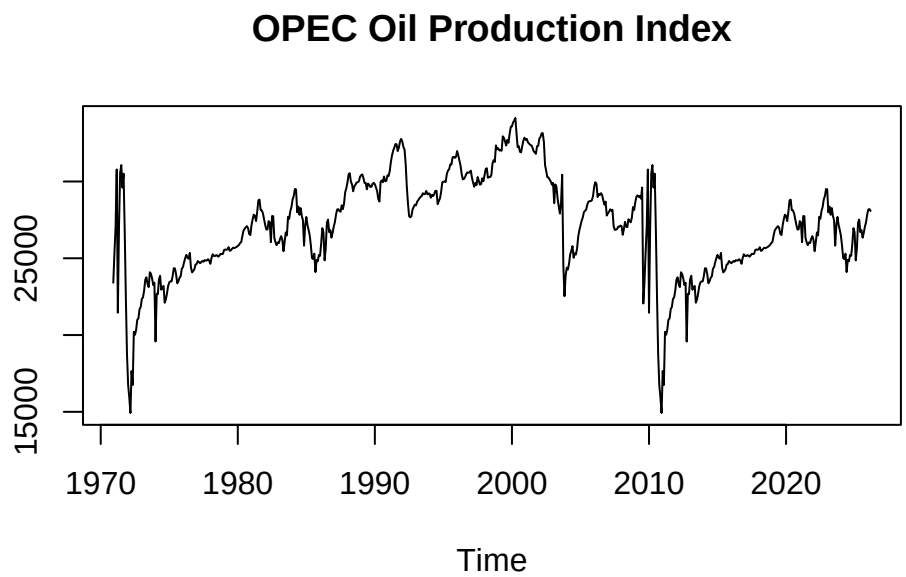
```r
plot(usd_idx_ts, main="U.S. Dollar Index", ylab="")
```




```r
plot(ind_prod_ts, main="Industrial Production Index", ylab="")
```



```
plot(opec_prod_ts, main="OPEC Oil Production Index", ylab="")
```



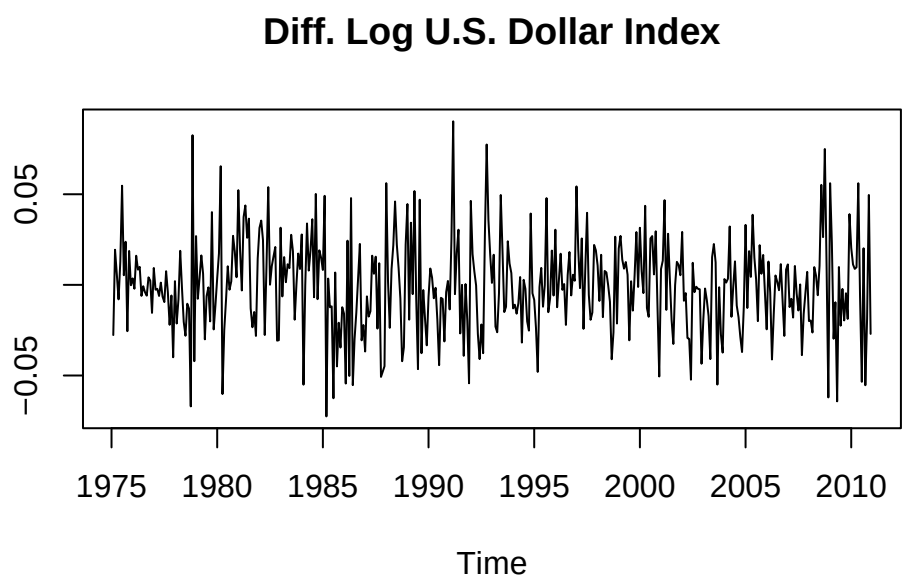
Transform log-differencing

```
d_usd_ts <- diff(usd_idx_ts)
dl_usd_idx_ts <- diff(log(usd_idx_ts))
dl_usd_idx_train_ts <- diff(log(usd_idx_train_ts))

#write.table(d_usd_ts, "savedData/d_usd_ts_new.csv", sep = ",", row.names = FALSE)
#write.table(dl_usd_idx_train_ts, "savedData/dl_usd_idx_train_ts_new.csv", sep = ",", row.nam


d_ind_ts <- diff(ind_prod_ts)
dl_ind_ts <- diff(log(ind_prod_ts))
dl_ind_train_ts <- diff(log(ind_prod_train_ts))

#write.table(d_ind_ts, "savedData/d_ind_ts_new.csv", sep = ",", row.names = FALSE)
#write.table(dl_ind_train_ts, "savedData/dl_ind_train_ts_new.csv", sep = ",", row.names = FAL
```

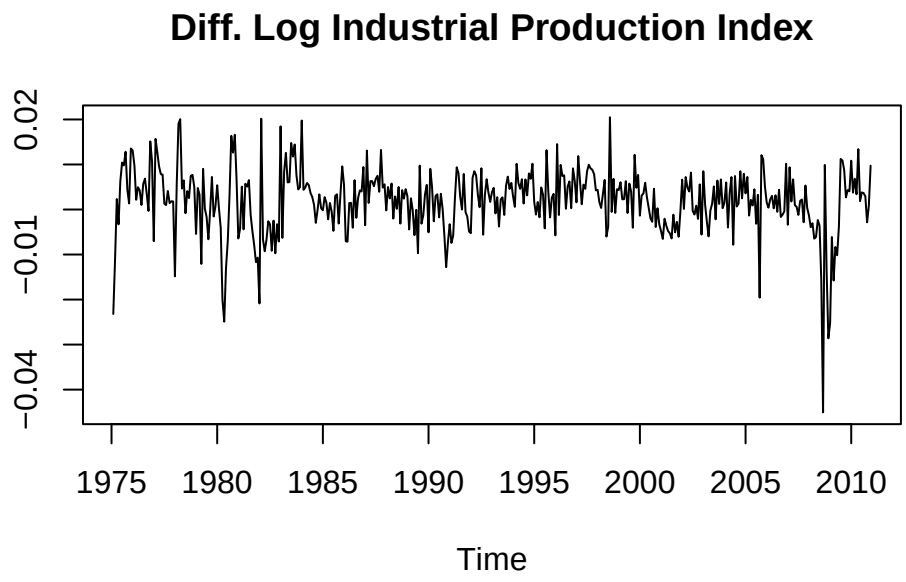
```
d_opec_ts <- diff(opec_prod_ts)
dl_opec_prod_ts <- diff(log(opec_prod_ts))
dl_opec_idx_train_ts <- diff(log(opec_prod_train_ts))

#write.table(d_opec_ts, "savedData/d_opec_ts_new.csv", sep = ",", row.names = FALSE)
#write.table(dl_opec_idx_train_ts, "savedData/dl_opec_idx_train_ts_new.csv", sep = ",", row.
```

Plot

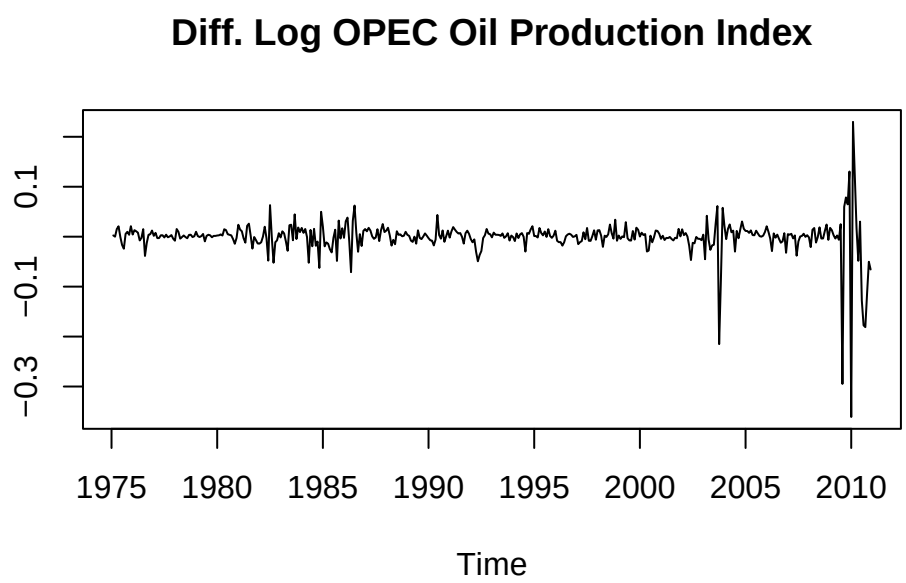
```
#plot(dl_usd_idx_train_ts, main="Difference U.S. Dollar Index", ylab="")
plot(dl_usd_idx_train_ts, main="Diff. Log U.S. Dollar Index", ylab="")
```




```
#plot(d_ind_train_ts, main="Difference Industrial Production Index", ylab="")
plot(dl_ind_train_ts, main="Diff. Log Industrial Production Index", ylab="")
```

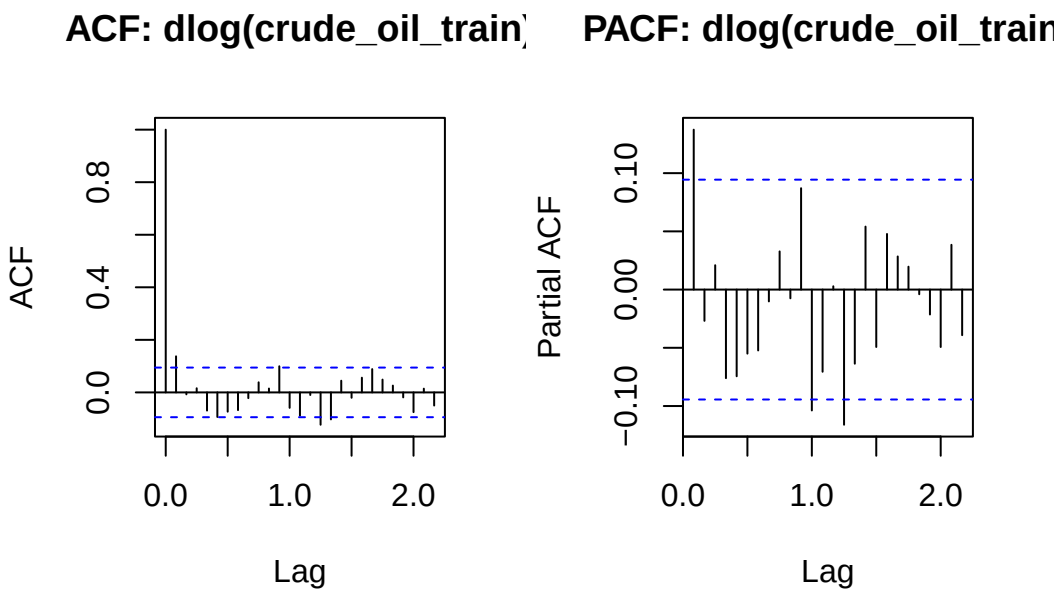



```
#plot(d_opec_idx_train_ts, main="Difference Industrial Production Index", ylab="")
plot(dl_opec_idx_train_ts, main="Diff. Log OPEC Oil Production Index", ylab="")
```



Diagnostics: variogram, ACF, PACF

```
par(mfrow=c(1,2))

acf(dl_crude_oil_train_ts,
    main = "ACF: dlog(crude_oil_train)")

pacf(dl_crude_oil_train_ts,
     main = "PACF: dlog(crude_oil_train)")
```

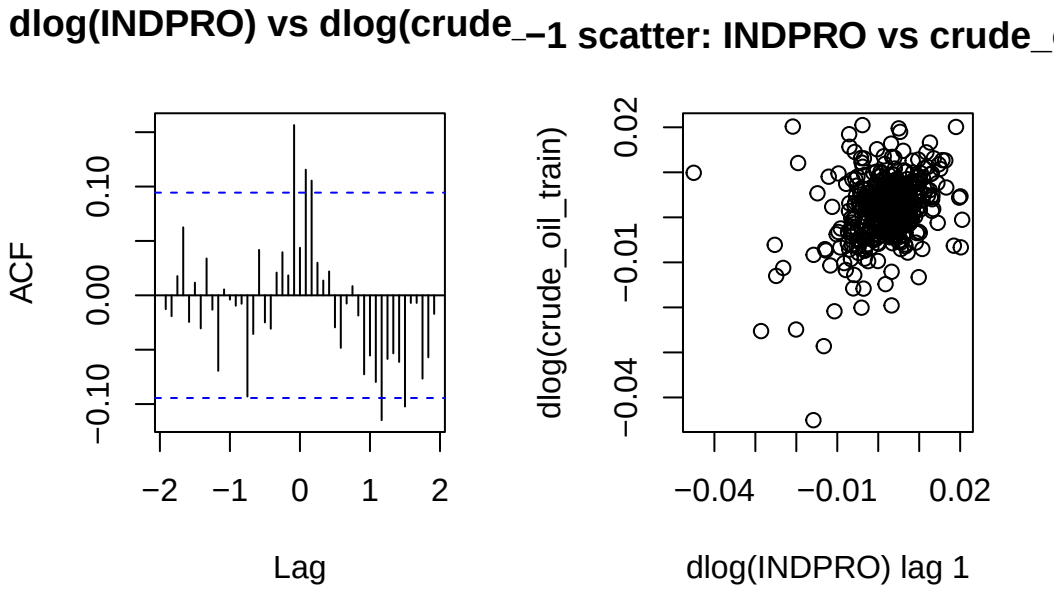


ARMAX

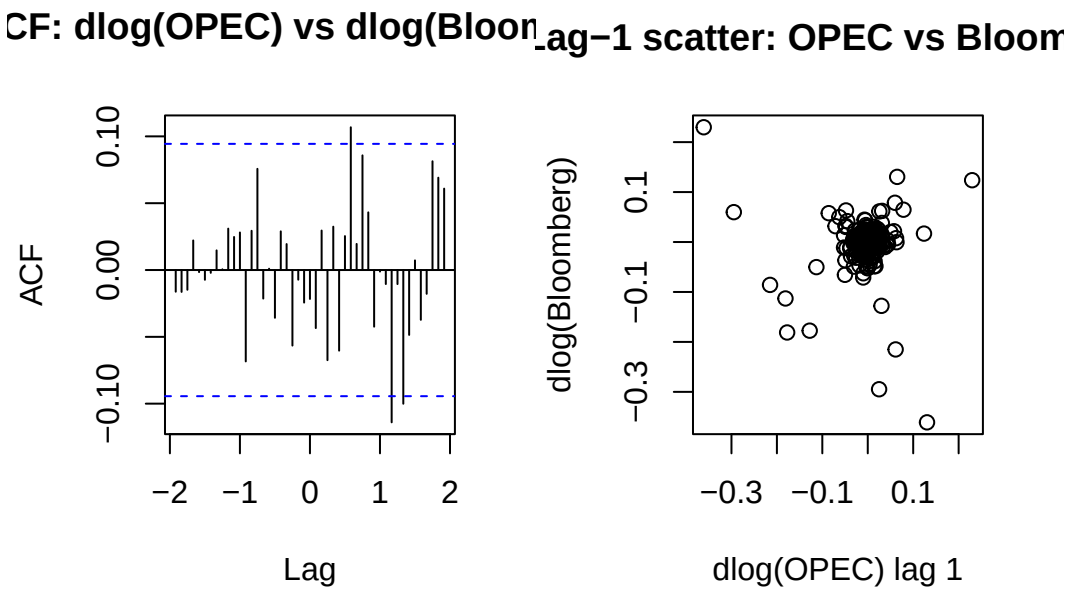
Cross-correlation diagnostics for leading variables

```
par(mfrow=c(1,2))

ccf(dl_ind_train_ts, dl_crude_oil_train_ts, main="CCF: dlog(INDPRO) vs dlog(crude_oil_train)"

plot(stats::lag(dl_ind_train_ts, -1), dl_ind_train_ts,
```

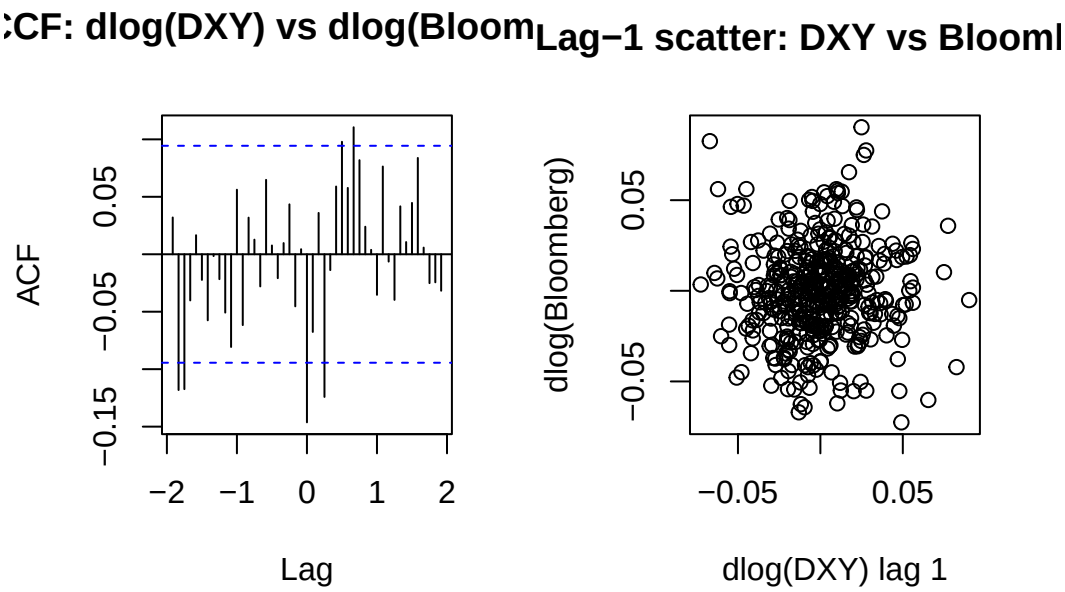
```
     xlab="dlog(INDPRO) lag 1", ylab="dlog(crude_oil_train)",
     main="Lag-1 scatter: INDPRO vs crude_oil_train")
```




```
par(mfrow=c(1,2))

ccf(dl_opec_idx_train_ts, dl_crude_oil_train_ts, main="CCF: dlog(OPEC) vs dlog(Bloomberg)")

plot(stats::lag(dl_opec_idx_train_ts, -1), dl_opec_idx_train_ts,
     xlab="dlog(OPEC) lag 1", ylab="dlog(Bloomberg)",
     main="Lag-1 scatter: OPEC vs Bloomberg")
```




```
par(mfrow=c(1,2))

ccf(dl_usd_idx_train_ts, dl_crude_oil_train_ts, main="CCF: dlog(DXY) vs dlog(Bloomberg)")

plot(stats::lag(dl_usd_idx_train_ts, -1), dl_usd_idx_train_ts,
     xlab="dlog(DXY) lag 1", ylab="dlog(Bloomberg)",
     main="Lag-1 scatter: DXY vs Bloomberg")
```



Align data

```
df_all <- ts.intersect(
  oil  = dl_crude_oil_ts,
  ind  = dl_ind_ts,
  opec = dl_opec_prod_ts,
  usd  = dl_usd_idx_ts
)

start(df_all)
```

```
[1] 1971    1
```

```
end(df_all)
```

```
[1] 2026    2
```

Split Train/Holdout

```
train_all <- window(df_all, start = c(1975, 2), end = c(2010, 12))
holdout_all <- window(df_all, start = c(2011, 1), end = c(2026, 1))

y_train <- train_all[, "oil"]
y_holdout <- holdout_all[, "oil"]

xreg_train <- as.matrix(train_all[, c("ind", "opec", "usd")])
xreg_holdout <- as.matrix(holdout_all[, c("ind", "opec", "usd")])
```

ARIMA baseline

```
# ARIMA baseline
arima_model <- auto.arima(
  y_train,
  seasonal = FALSE,
  stationary = TRUE
)

print(arima_model)
```

```
Series: y_train
ARIMA(0,0,1) with zero mean

Coefficients:
         ma1
      0.1459
s.e.  0.0486

sigma^2 = 0.007729:  log likelihood = 436.85
AIC=-869.7   AICc=-869.68   BIC=-861.57
```

```
arima_res <- arima_rolling_fc(arima_model, y_holdout)
```

ARMAX candidate model

```r
armax_model <- auto.arima(
  y_train,
  xreg = xreg_train,
  seasonal = FALSE,
  stationary = TRUE
)

print(armax_model)
```

```
Series: y_train
Regression with ARIMA(0,0,1) errors

Coefficients:
         ma1     ind     opec      usd
      0.1301  0.3711  -0.0250  -0.4817
s.e.  0.0505  0.6032   0.1128   0.1609

sigma^2 = 0.007621:  log likelihood = 441.4
AIC=-872.8   AICc=-872.66   BIC=-852.47
```

extract nonseasonal ARMA order

```r
order_armax <- c(armax_model$arma[1], 0, armax_model$arma[2])
include_mean_armax <- any(names(armax_model$coef) %in% c("intercept", "mean"))
```

Rolling 1-step ARMAX

```r
armax_res <- rolling_armax_fixed(
  y_train = y_train,
  x_train = xreg_train,
  y_holdout = y_holdout,
  x_holdout = xreg_holdout,
  order = order_armax,
  include_mean = include_mean_armax
)
```

RMSE comparison

```r
cat("ARIMA RMSE:", arima_res$rmse, "\n")
```

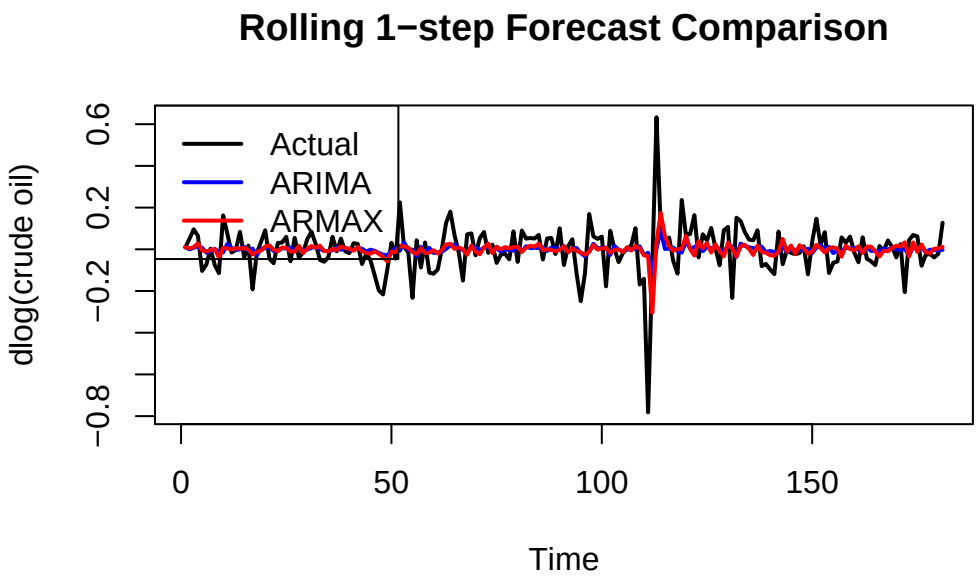
```
ARIMA RMSE: 0.1131622
```

```r
cat("ARMAX RMSE:", armax_res$rmse, "\n")
```

```
ARMAX RMSE: 0.1117057
```

Plot forecasts

```r
plot(as.numeric(y_holdout), type = "l", col = "black", lwd = 2,
     main = "Rolling 1-step Forecast Comparison",
     ylab = "dlog(crude oil)", xlab = "Time")

lines(as.numeric(arima_res$fc), col = "blue", lwd = 2)
lines(as.numeric(armax_res$fc), col = "red", lwd = 2)

legend("topleft",
       legend = c("Actual", "ARIMA", "ARMAX"),
       col = c("black", "blue", "red"),
       lty = 1, lwd = 2)
```



Simple Rules

TODO